

Digital Twins for Predictive Maintenance in Automotive Aftersales

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Abstract—TODO: Abstract here.

Index Terms—Digital Twin, Predictive Maintenance, Industry 4.0, Machine Learning, IoT, Condition Monitoring

I. INTRODUCTION

- Hook
- Smart to Wise Bezug
- Problemstellung
- Zeilsetzung
- RQ, Forschungslücke
- Beitrag der Arbeit
- Aufbau der Arbeit

II. RELATED WORKS & THEORETICAL BACKGROUND

A. Digital Twins

B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [1]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [1], [2]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [2]; and predictive maintenance (PdM) extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until end-of-life is reached [1], [3].

PdM relies on three data layers: on-board sensors delivering high-frequency signals such as vibration, current, voltage, and cell temperature [2]–[4]; the On-Board Diagnostic (OBD) interface providing standardized diagnostic codes [5], [6]; and over-the-air (OTA) telematics enabling continuous low-latency exchange with the backend [4], [6], [7]. Together, these layers transform the vehicle from a periodically inspected asset into a continuously observed system, a shift first documented by Zhang et al. [5] in their work on Connected Vehicle Diagnostics and Prognostics at General Motors.

PdM at this scale is not a stand-alone technique but a service layer built on a digital twin (DT) of the vehicle. As Lopes et al. [7] and Hadraoui et al. [2] describe, the DT supplies the synchronized virtual representation against which sensor streams are interpreted, anomalies detected, and remaining-useful-life estimates computed. Without a DT, telemetry must be processed in isolation; without PdM, the DT remains a

passive monitor. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [4], [7].

C. Machine Learning in PdM

Three problem classes structure ML-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks on time-frequency representations of vibration signals [1], [3]. Regression targets Remaining Useful Life (RUL) estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [1], [4]. Anomaly detection operates without explicit labels, learning nominal behavior and flagging deviations as failure precursors [1], [4]. AbouGrad et al. [1] report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they emphasize that these figures depend strongly on dataset quality and operating conditions.

A recurring constraint is data: AbouGrad et al. [1] and Hadraoui et al. [2] note that preprocessing consumes 70–90% of project time, genuine failure data is structurally scarce, and public benchmarks representative of in-service behavior are largely absent. Model-based and hybrid approaches that embed physical degradation laws into the learning pipeline are therefore preferred over purely data-driven ones, as they extrapolate more reliably under data scarcity [2], [3], [8]. Beyond accuracy, deployment in safety-relevant contexts raises the question of explainability: deep architectures achieve high accuracy but operate as opaque models whose decisions cannot readily be justified to engineers, customers, or regulators [1]. Explainable AI (XAI) techniques such as feature attribution and attention-based saliency are increasingly integrated into PdM pipelines to expose the reasoning behind a prediction, marking the conceptual bridge from smart to wise information systems [1].

D. Automotive Aftersales & Customer Lifecycle

Forschungslücke

While these contributions establish the technical foundations of DT-based PdM, existing work concentrates either on

algorithmic aspects in isolation [1], [3] or on industrial and aerospace applications [8]. The integration of DT-based PdM into automotive aftersales business models — including its implications for customer interaction, upselling potential, and vehicle lifecycle value — remains underexplored. The present work addresses this gap.

III. METHODOLOGY

- LR
- Illustrative Case Study
- (DSR)

IV. REFERENCE ARCHITECTURE

- Sensorik
- ML-Algorithmen
- Kennzahlen
- Schnittstellen
- Datenmanagement

V. ILLUSTRATIVE CASE STUDY

- A. Lifecycle Comparison
- B. Assumption Framework
- C. Illustrative Calculation & Results
- D. Limitations

VI. DISCUSSION AND RECOMMENDATIONS

- Interpretation der Ergebnisse
- Implikationen fuer Forschung und Praxis

VII. CONCLUSION & FUTURE WORK

- Zusammenfassung der wichtigsten Erkenntnisse
- Beitrag der Arbeit
- Ausblick auf weitere Forschung [?]

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ERKLÄRUNG ZUR NUTZUNG VON KI-TOOLS

TODO: KI-Erklärung hier einfügen (gemäß DHBW-Vorgabe).